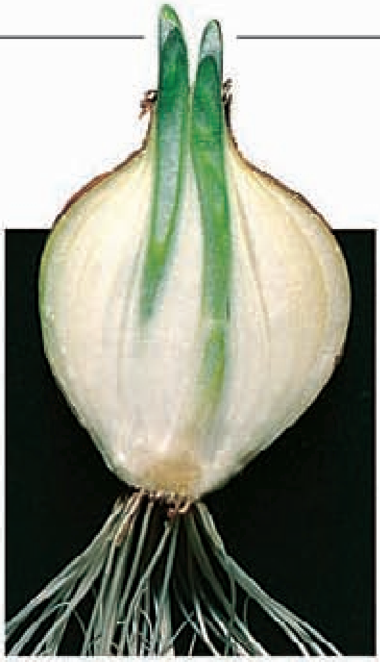
Chloroplasts in cells collect sunlight

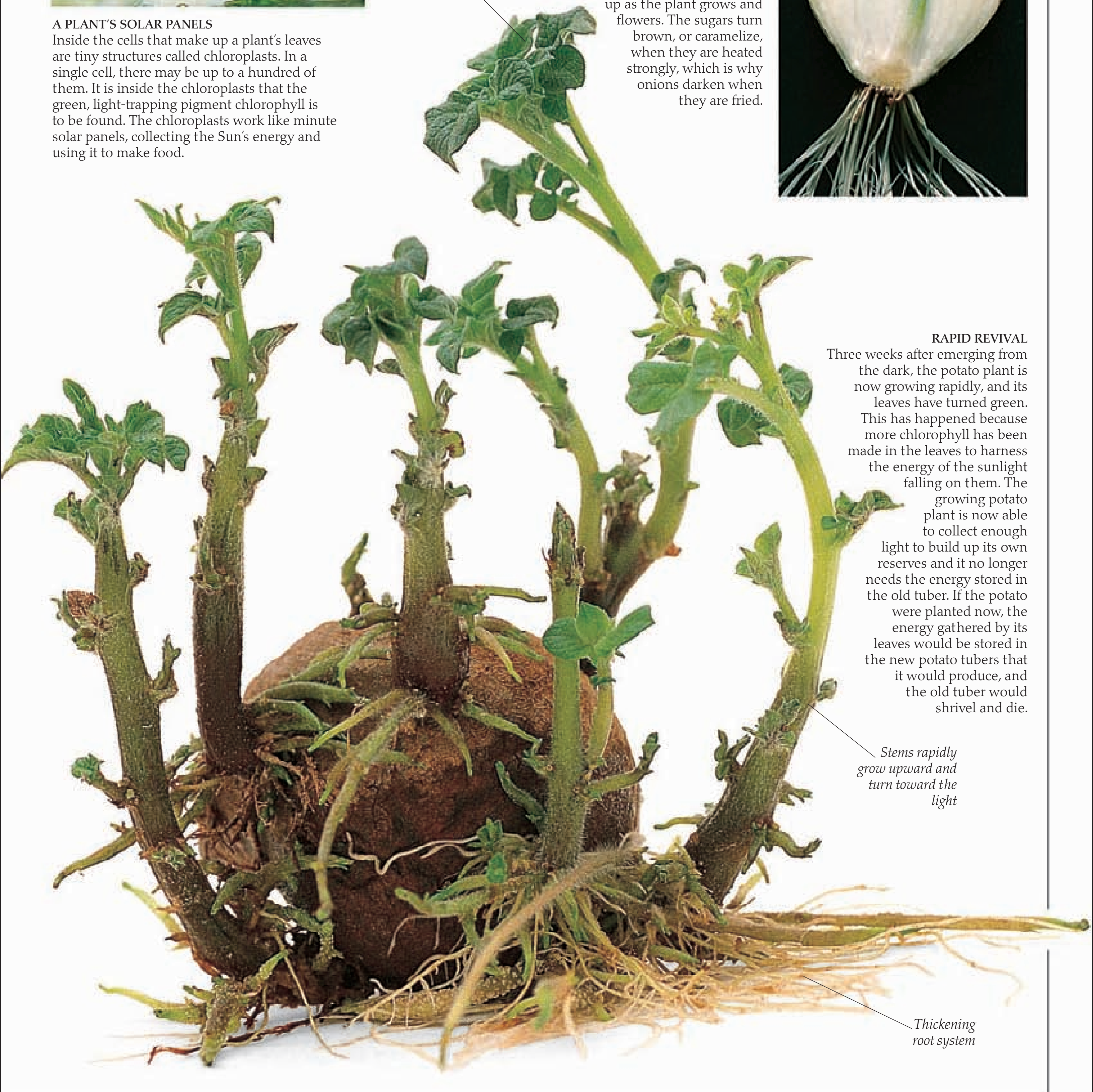
A PLANT'S SOLAR PANELS

Inside the cells that make up a plant's leaves are tiny structures called chloroplasts. In a single cell, there may be up to a hundred of them. It is inside the chloroplasts that the green, light-trapping pigment chlorophyll is to be found. The chloroplasts work like minute solar panels, collecting the Sun's energy and using it to make food.

STORING SUGAR

Plants store food in various ways—as starches, sugars, or oils. In its first year, an onion plant stores sugars in the onion bulb, which is made up of swollen leaf bases around a shortened stem. In the second year, the sugars in the onion bulb are used up as the plant grows and flowers. The sugars turn brown, or caramelize, when they are heated strongly, which is why onions darken when they are fried.





Green leaves rich in chlorophyll

RAPID REVIVAL

Three weeks after emerging from the dark, the potato plant is now growing rapidly, and its leaves have turned green. This has happened because more chlorophyll has been made in the leaves to harness the energy of the sunlight falling on them. The growing potato plant is now able to collect enough light to build up its own reserves and it no longer needs the energy stored in the old tuber. If the potato were planted now, the energy gathered by its leaves would be stored in the new potato tubers that it would produce, and the old tuber would shrivel and die.

Stems rapidly grow upward and turn toward the light

Thickening root system